

Fable Reborn: Investigating Gameplay Experience between a Human Player and a Virtual Dungeon Master

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ABSTRACT

Dungeons & Dragons (D&D) is a role-playing game typically played face-to-face with participants seated around a tabletop. Online technologies have enabled the conduct of live sessions on virtual platforms, allowing players to continue enjoying the game despite geographic distances. Playing D&D relies heavily on a dungeon master who is tasked with preparing the fictional world and facilitating the interaction between players who each control a single story character. In this paper, we investigate the impact of employing AI techniques in designing a virtual dungeon master (VDM) on the gameplay experience of D&D players. The VDM utilizes Open AI's GPT2 model to generate story prompts in combination with an input plausibility checker, a dice rolling feature, and a candidate story prompt scorer. Feedback from D&D players revealed that the VDM is capable of producing believable fantasy stories but lacks consistency in the quality of its generated stories. Our findings point to the potential abilities of an artificially intelligent agent in engaging human D&D players in role-playing games; however, the VDM is still not capable of truly replicating the experience of playing with a human DM.

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces**: *Interaction paradigms*; *Interaction design*.

KEYWORDS

role-playing game, Dungeons & Dragons, virtual dungeon master, interactive storytelling, gameplay experience

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1 INTRODUCTION

Role-playing games (RPG) engage the players by enabling them to take control of a character's actions, thoughts, and dialogues with the other players, and how they behave in the fictional game world. A form of RPG that can accommodate a higher degree of interactivity from its players is the tabletop role-playing game (TTRPG). This is evident in Dungeons & Dragons (D&D), a popular TTRPG that is typically played face-to-face with participants seated around a tabletop. Online technologies have enabled the conduct of live sessions of D&D on virtual platforms [8], allowing players to continue enjoying the game despite geographic distances.

Playing D&D has always been constrained with the requirement to have one of the players be the dungeon master (DM). The DM prepares the fictional world pre-game and facilitates the narration of the story in-game together with the players who role-play and control the actions of their own characters. The DM functions both as a moderator and a storyteller for the adventures that the players will role-play. Preparing the story world before a game requires conceptualizing the game world and formulating the adventures that players or story characters will experience. During the game, the DM tracks all the details of the story, including the story world state and the conditions of the characters that change dynamically in the story world.

An interactive storytelling environment can be employed to facilitate the conduct of role-playing games such as D&D. Interactive storytelling is a form of automated story generation that takes user inputs into account in the creation of story events, allowing users to have an interactive experience as their choices influence the progression and eventual outcome of the story [5, 9]. This is usually executed by allowing users to assume the role of a character performing actions and affecting events in a fictional world [14]. Having the users believe that they are important and integral to the story and that their actions have a meaningful impact on the sequence and outcome of events are the goals of interactive storytelling [10]. This may aid in increasing their engagement and enjoyment with the story.

The number of tasks expected from a DM before and during a game, the instances when a human DM is unavailable to lead a gameplay session, and advancements in natural language processing motivated the development of a virtual dungeon master. While a virtual DM can be seen as beneficial to the conduct of a game, having a human DM may lead to differing game dynamics and may cause variances in the players' decisions than with a virtual DM. A bias may arise with a human DM, where they could be forgiving to

the players, while a virtual DM will have machine-generated decisions and limited knowledge base from which to draw its narrative content. The game experience itself will be different when playing on a tabletop where all the players are in the same physical space, and when playing online with the players interacting through a text-based interface in geographically separate locations.

In this paper, we describe Fable Reborn, an interactive storytelling environment that employs a virtual DM (VDM) in the production of fictional stories by adapting rules from the D&D 5th edition (5e) playbook. Fable Reborn has a number of parallel features that are present in **AI Dungeon**¹. Both systems allow multiplayer sessions and use OpenAI's GPT-2 model for story generation. However, AI Dungeon's story generation is "infinite"; story text will continue to be generated as long as the players still have input. Fable Reborn, on the other hand, incorporates the concept of a goal that players should strive to reach to end the game. This goal is presented as the starting prompt of each game session. Fable Reborn also utilizes other components to improve its story generation capabilities by checking the plausibility of a player's input, imitating D&D's dice roll to determine an input's success rate, and ranking the generated story text based on their alignment to the context of the current story.

Feedback from D&D players showed that the VDM is capable of producing engaging and believable fantasy stories but lacks consistency in the quality of its generated stories. The VDM is also able to formulate responses that are aligned with the human players' story text inputs while occasionally introducing random and unrelated new elements. The inconsistency and misalignment in the VDM's responses affected the player's gameplay experience. These findings point to the potential abilities of Fable Reborn to provide an alternative platform for D&D players to engage in role-playing games but is still not capable of truly replicating the experience of playing with a human DM.

The rest of the paper is organized as follows. In Section 2, we give an overview of the rules and mechanics of D&D 5e that informed the design of Fable Reborn as well as the other features of the interactive storytelling environment. This is followed by the methodology and the results from user evaluation on their gameplay experience in interacting with the virtual DM in Sections 3 and 4, respectively. We summarize our findings and suggestions on migrating tabletop games to digital platforms with the use of AI and web-based technologies in Section 5.

2 THE VIRTUAL DUNGEON MASTER

The design of Fable Reborn focused on adapting the D&D 5th edition (5e) rules from only the *Player's Handbook* by Wizards of the Coast. The decision to follow 5e is based on a report by the Orr Group in 2021 [8] which showed that more than 50% of their total played campaigns in the 4th quarter of 2020 was D&D 5e campaigns.

2.1 D&D Rules and Mechanics

D&D has many editions of rules and mechanics with the 5th edition being the latest ruleset released in 2014. The basic rules are as follows:

- A DM creates and narrates the story, and adjusts the story based on the actions of the characters (formulated by the players) and their interaction with other characters/players, non-playing characters (NPCs).
- Guided by the D&D character creation mechanics, players role-play their own characters that they have defined at the start of the game.
- Multiple dice are used to determine the success of the players' actions. Each die has a specific usage, for example, a 20-sided die is usually used to determine skill effectiveness level while a 6-sided die is used to determine the combat effects and candidate actions.

Two (2) key figures are integral to the D&D game: the dungeon master and the players. The DMs exert the most effort pre-game as they are tasked with creating worlds for other players to explore and adventures that drive the story. The rules allow the dungeon master to balance his/her power while maximizing the fun in the game. Rules are also used to guide the DM in handling events as they occur in the story world. Players, on the other hand, follow a character creation process by choosing a race, character class, background, and the distribution of their character's abilities. Other character details such as name, age, gender, and alignment are also chosen. Figure 1 illustrates Fable Reborn's Character Sheet and the corresponding User Profile page in Figure 2. All of these attributes are used in-game to guide a player's role-playing, determine if a player's action can be executed in the story world and how it may affect other players, and influence the flow of story events.

D&D games usually start off with the DM generating a campaign that represents the quest a player must achieve. The DM explains the setting; this is followed by the players introducing the characters they are role-playing or discussing the course of action to be taken by the group [13]. The DM then determines the encounters the group runs into such as main events or combat events. Players issue actions or commands to advance in the game. The players can virtually do anything with their characters as long as the DM allows them to and subject to the constraints dictated by the rules.

Dice rolls are the main deciding factor of the game. It is used in many different scenarios, primarily to determine the probability of the success of a character's actions and the magnitude of an attack committed to an entity during combat, especially when it is a particularly difficult action to be judged solely by the DM. Determining whether a player's attack gravely wounds someone, the success of infiltrating a secret camp, and breaking down a door can be decided by dice rolls. The dices to be used in the D&D games are determined by the number of sides it contains, with the 20-sided dice being used more often.

2.2 Fable Reborn

Playing D&D heavily relies on the usage of words to describe the events that are taking place and the actions that a character may want to perform in the game world. The virtual DM uses words to describe the story world and the outcome of player actions as vividly as possible to enable the players to recreate the world as best as they can using their own imagination. Players also use words to express the actions of the characters they are role-playing.

¹<https://play.aidungeon.io/main/home>

Figure 1: Character Sheet

Figure 2: User Profile

Fable Reborn follows a six-step process in the generation of stories as shown in Figure 3. A dialogue takes place when the player responds to the story prompt generated by VDM. An **input**

plausibility checker validates if the player input is an action they can be performed in the virtual story world given the current progress of the story. Table 1 illustrates an instance when the virtual DM tag the input from player *Scarlet* as implausible because the two nouns, *Scarlet* and *Juan*, are not present in the previous story prompts. If the player were to change the text into “*Scarlet wakes Juan up in the Outpost Settlement*” or “*Scarlet wakes Juan up to go and get their meals*” which contains the nouns “*Outpost Settlement*” or “*meals*”, then the VDM would have accepted the input.

Once a player input is accepted, the VDM performs a dice roll to determine the success rate of the player’s action. The **dice roll** feature mimics a 20-sided die roll to simulate ability checks in D&D 5e games. This is typically done whenever a player performs an action that requires a certain skill. The value of the dice is added to the relevant character ability as a bonus. The resulting number must reach a certain threshold called a *difficulty class* in D&D 5e in order for the player’s action not to fail.

The player input is then combined with the previous story prompts to form a single text string and cleaned to remove unnecessary characters. The string is encoded to **OpenAI’s GPT-2 transformer model**² that was trained with fantasy stories from **chooseyourstory.com** and fine-tuned following the hyperparameters of **AI Dungeon** in order to generate three (3) candidate story prompts. The candidate story prompts are pitted against the player input and the existing story using a **scorer** to determine which among the three is the most related to the the overall story. The scorer uses the automatic evaluation metric model for text generation, **BERTScore** [15], that computes for token similarity using contextual embeddings.

Once the VDM determines the best candidate story text, it displays this on the interface, shown in Figure 4, and the whole story generation process repeats again. The user interface (UI) is a critical component of the game as it affects the player-computer interaction. A minimalistic UI that supports text-based interaction is made available for the players to communicate with the virtual DM. It is also the means by which the virtual DM can present various information to the players. The *In the Dungeon* space lists the players who are role-playing their characters, while a separate *Chat* box is provided to allow players to exchange messages with one another.

3 METHOD

Fable Reborn is designed as an online multiplayer interactive environment targeted towards an audience that is familiar with D&D. It should function similarly to how regular tabletop D&D games should be played where there is a DM who prepares the campaign

²<https://openai.com/blog/tags/gpt-2/>

Table 1: Sample excerpt illustrating how the VDM detects and responds to implausible player input.

DM:	You find yourselves in the Outpost Settlement as you wake up at sunrise. You see people emerging to wash and prepare their meals as the day guards takes over the shift.
Scarlet:	Scarlet wakes Juan up.
DM:	Player input not plausible. Please enter a new action.

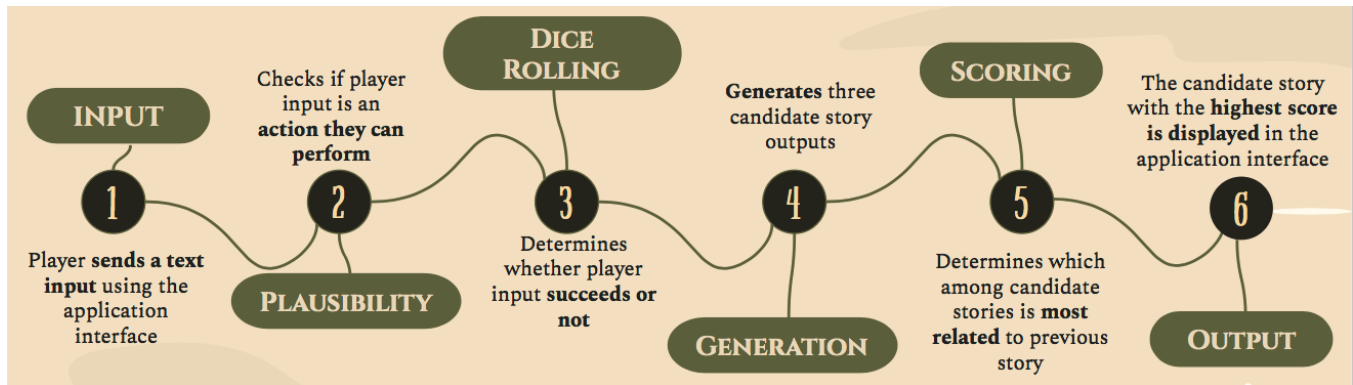


Figure 3: Story generation pipeline of Fable Reborn

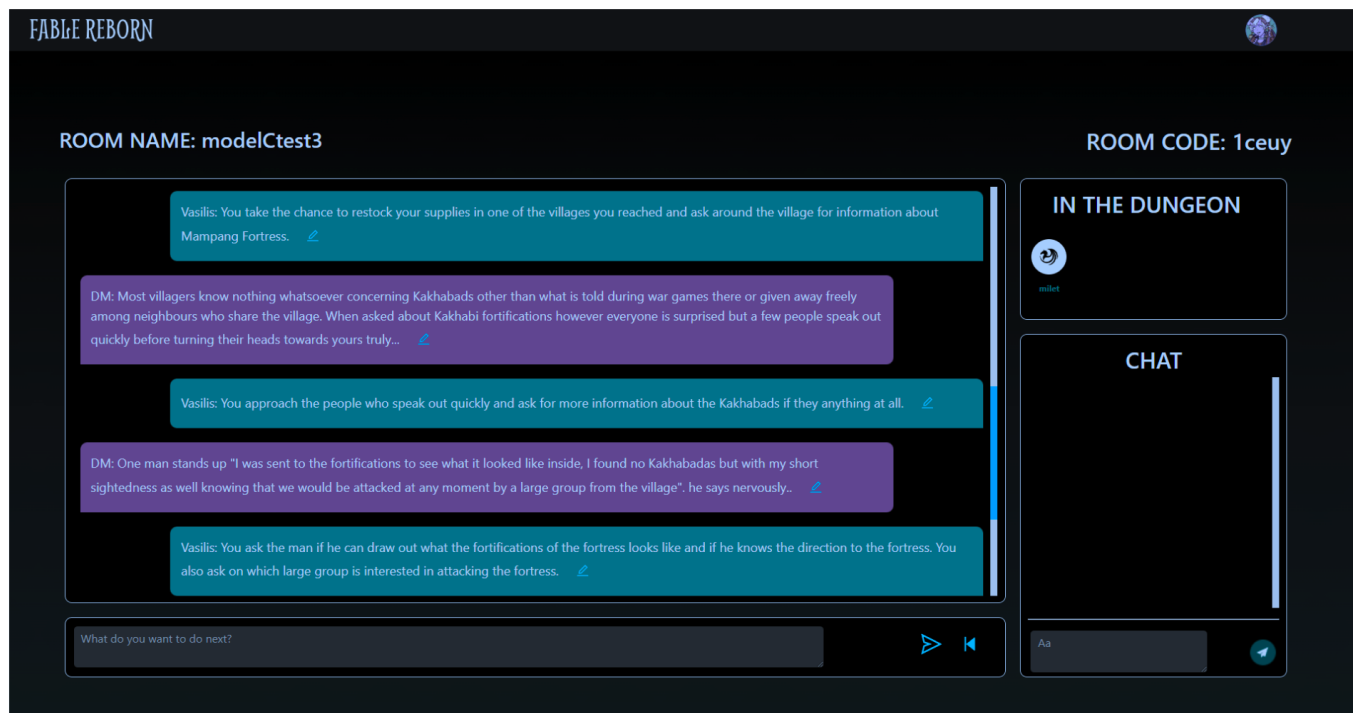


Figure 4: Fable Reborn virtual campaign game room

and narrates the story based on the player's defined characters and their actions in the story world. However, instead of playing with a human DM, Fable Reborn introduces a virtual DM to facilitate the storytelling feature of the game. In this section, we discuss the method we employed in evaluating the gameplay experience arising with the shift from face-to-face verbal to online text-based interaction and from human to a virtual DM.

30 participants who are D&D players and/or dungeon masters participated in the play testing of Fable Reborn. They were formed into 11 groups according to their availability. Before the actual play testing, the players accomplished *informed consent forms* and were given an orientation on how to navigate the system and the commands they can use.

At the start of the game session, players were required to create their characters for their campaigns using the online character sheet shown in Figure 1 and to select a campaign (game) room. Players were given free reign on the responses they want to provide to the VDM based on the latter's generated stories. Each group of players had to test the system for at least 30 minutes. This constraint was put in place due to the high computing power needed to run Fable Reborn, specifically the GPT2 model for story generation and the BERTScore candidate story text scorer.

When the play testing session ended, players individually accomplished a post-test evaluation survey to give their feedback on the performance of the virtual DM and to share their experience

Table 2: Session details for the 11 playtests.

Session	Pax	Duration	Human DM
1	4	32.46	3
2	4	68.30	0
3	4	37.02	3
4	2	46.04	1
5	2	39.23	1
6	1	36.43	0
7	2	58.05	2
8	3	40.38	1
9	2	57.30	1
10	1	40.43	0
11	5	49.24	0

in playing Fable Reborn. Three sets of criteria were used in evaluating Fable Reborn: the *narrative content* is evaluated following the StoryEval framework [11]; the *gameplay* is evaluated following the Game Experience Questionnaire [6]; and the *User Interface* is evaluated following the criteria in the Standardized Usability Scale [2]. Evaluation results for gameplay and user interface are the focus of our analysis and discussion in the following sections.

4 RESULTS

The change in interaction mode, from a face-to-face verbal interface to a web-based text interface, as well as the presence of an intelligent artificial entity, the VDM, may have affected the gameplay experience of D&D players. In this section, we present the results from players' evaluation of playing with a VDM and interacting through the user interface of Fable Reborn.

Table 2 shows the play testing details for the 11 sessions. All participants have tabletop D&D experience. 7 of the game sessions have at least 1 participant who has served as a human DM. Each session had an average play time of 45 minutes for the campaigns but none of the campaigns reached a story prompt that contained the campaign goal.

4.1 Gameplay Evaluation

We utilized the *Game Experience Questionnaire* [6] to probe the player's perception on how the change in interaction mode and the

Table 3: Gameplay Experience average evaluation scores using a linear scale of 0 to 4 with 0 representing not at all, 1 representing slightly, 2 representing moderately, 3 representing fairly, and 4 representing extremely.

Component	Average Score
Competence	1.88
Sensory and Imaginative Immersion	2.86
Flow	1.55
Tension and Annoyance	0.66
Challenge	1.04
Negative Affect	1.20
Positive Affect	2.95

Table 4: Sample excerpt from play session 9.

DM:	Last night's journey was all silent. You wonder if you have passed Kakhabads, or if your journey is just that dangerous.
Player:	Frank sleeps.

use of an intelligent entity affected their experience in playing D&D. The questionnaire has three (3) parts: core questionnaire, social presence, and post-game evaluation. The core questionnaire contains seven (7) components: competence, sensory and imaginative immersion, flow, tension or annoyance, challenge, negative affect, and positive affect. It uses a linear scale of 0 to 4 with 0 representing not at all, 1 representing slightly, 2 representing moderately, 3 representing fairly, and 4 representing extremely. Table 3 shows the average scores from the players' evaluation.

Competence measures how skillful participants felt during the game and is related to accomplishment and pride [7]. With an average score of 1.88, participants did not feel a sense of accomplishment in using text-based input to role-play their characters as opposed to verbally delivering their characters' thoughts and actions, often accompanied by gestures and tone. Verbal communication differs from written communication in that the former is designed to be simpler and more repetitive. As seen in the game log excerpt in Table 4, the player input is stated in a simple sentence without any other details. If conveyed verbally, physical expression can be used to assume that the character sets up camp or bed roll where they might sleep. In the absence of gestures and tone in written communication, the input can be taken in any number of ways depending on who is interpreting it, such as the character falling asleep instantly while on the journey or lying down on the road and falling asleep.

Sensory and imaginative immersion refers to the cognitive state of being in the game where the players' senses are placed in another reality [3]. It can be measured by the VDM's ability to bring about a feeling of being part of the game. The average score of 2.86 reflects players' mixed sentiment in VDM's ability to generate a story that is from the fantasy genre with fictional and unrealistic events. A player described this as VDM's ability to "produce a coherent story and used a singular theme". However, VDM also has a tendency to "take them out of the immersion because it would have some wrong grammar, misspell character names, and introduce story elements that were out of context".

Flow is a state of intense concentration and it got an average rating of 1.55. The presence of the transformer model and the scorer model, both of which are computing intensive, caused delays in VDM's generation of story prompts which can disrupt a player's concentration and momentum with the game. This observation is also reflected in *tension* which evaluates players' feelings of annoyance, irritation or frustration when the VDM takes an average of two to three minutes to generate a response. With the short time for play testing and none of the campaigns reached their goal, the score for *challenge*, 1.04, was affected as well.

Games are known to be inducers of affect because of the emotional interaction between the game and the players [4]. Fable Reborn received an average rating of 1.20 for *negative affect* which was caused by the speed of the story prompt generation where

Table 5: Social Presence average evaluation scores using a linear scale of 0 to 4 with 0 representing not at all, 1 representing slightly, 2 representing moderately, 3 representing fairly, and 4 representing extremely.

Social Presence	Score
Empathy	3.11
Negative Feelings	1.52
Behavioral Involvement	3.00

the participants felt slight boredom and found this process to be tiresome. On the other hand, the *positive affect* component got an average rating of 2.95 which reflects the participants' enjoyment during their time playing Fable Reborn and were impressed with the responses of the virtual DM.

The **social presence** component evaluates how the participants influenced and affected each other's empathy, negative feelings, and behavioral involvement. According to [6], this questionnaire can be administered when co-players are either virtual or co-located. The average scores garnered by Fable Reborn are shown in Table 5. *Empathy*, with an average score of 3.11, highlights how participants were positively affected by others' actions and their empathy with each others' choices in directing the flow of the shared story. *Negative feelings* of jealousy, revenge or malicious delight are also evident with the average score of 1.52. This can be attributed to the variances in the quality of story prompts that the players received from the VDM. Moreover, VDM does not place any restrictions on the types of actions a player may formulate and freely accepts story input such as those that can cause harm to other characters. *Behavioral involvement* evaluates the relatedness of the players' actions. It got an average score of 3.00 which shows that the players are aware and understand that their actions affect others' actions and experience.

The last component of the Game Experience Questionnaire, the **post-game evaluation** is used to assess players' physical and mental states after they stopped playing D&D. The evaluation scores are shown in Table 6.

Positive experience received an average score of 2.21, indicating players feeling energized, satisfied, and revived after playing the game. Although participants admitted that they were enjoying the game at first, the story became confusing when the VDM started introducing random characters, which affected their positive experience with Fable Reborn. Regardless, a participant shared that "*the*

Table 6: Post-game Evaluation average scores using a linear scale of 0 to 4 with 0 representing not at all, 1 representing slightly, 2 representing moderately, 3 representing fairly, and 4 representing extremely.

Post-Game	Score
Positive Experience	2.21
Negative Experience	0.54
Tiredness	0.43
Returning to Reality	1.04

game may give weary (human) DMs a rest even though there is a lot more to be desired" on the performance of the virtual DM.

The *negative experience*, on the other hand, received an average score of 0.54. This is indicative of the absence of negative gameplay experience, such as feeling bad, guilty, exhausted, and wasted time, among the participants. Despite this, most participants raised their concern on the long time it took for the VDM to generate the next story text. A similar low average score of 0.43 was garnered in the *tiredness* attribute, indicating that the participants did not feel any exhaustion or weariness from playing Fable Reborn.

The time it takes to generate the next story text, which averages from two to three minutes, may have caused the average score of 1.04 in *returning to reality*. That is, the participants were not giving their full attention to the game; consequently, they were not as immersed with the fictional world and had no difficulty returning to reality. During the story generation process, it was observed that the participants were frequently talking with each other on topics that are not always associated with the game; some are even browsing their social media accounts while waiting for the VDM to display a response.

4.2 User Interface Evaluation

With the use of a web-based interface, the verbal interaction between players is transferred to a text-based interface. Thus, the design of Fable Reborn's user interface is integral in facilitating or constraining the interaction among players and with the virtual DM. The Standardized Usability Scale (SUS) [2] that contains 10 questions with a 5-point Likert scale corresponding to *strongly disagree* (1) to *strongly agree* (5) was used for this purpose.

Usability items on the ease and confidence in using Fable Reborn, the well-integrated functions provided by the system, and the willingness of the participants to play the game again received an average score of 4.114. The SUS score was also computed and evaluated following the adjectival scale [1] and interpreted using the adjectival rating of Sauro [12]. Fable Reborn received an SUS mean score of 76 out of 100 with a standard deviation of 12.64; this is equivalent to an adjectival rating of "good".

Despite the availability of existing messaging platforms that would allow players to exchange messages with one another in-game, it was observed that the players utilized the chat feature of Fable Reborn to talk about the actions they would like to perform individually before sending their input to the VDM. The players take their time deliberating on the best course of action and also share in the outcome of each other's actions. This is necessary since VDM can only process one player input at a time due to the computing power it requires in generating the next story text.

Overall feedback showed that participants found the design of Fable Reborn to be "*intuitive and the interface was easy enough to understand*" (P4). It should be reiterated that this assessment is the perception of players who are well-versed with the mechanics of a D&D game. Furthermore, players described Fable Reborn as a "*fun game*" despite the limitations of the story generation module in adding new elements to the story, but "*the website was easy to use and straightforward and the design looks great*" (P28).

Participants shared their suggestions on features that can benefit the players, such as a note taking component for the game room, a

character sheet where players can jot down notes for next steps or actions, and a space where they can monitor the current state of their co-players.

5 DISCUSSION

The use of a text-based interface combined with an artificially intelligent virtual DM may provide an alternative platform for players of D&D. However, the dynamics of the game may differ between a human-generated vs a machine-generated decision on how a player's input influences the progression of the story. We summarize our findings on the gameplay experience into two clusters: playing with a virtual DM and the use of text-based interaction.

Virtual DM. The performance of the VDM is primarily affected by its ability to generate interesting and challenging campaigns that can immerse players into the virtual world of the game. Moreover, story prompts that are aligned with the players' inputs can also increase enjoyment and engagement with the game. Participants found the use of a virtual DM to be a promising idea that may lead to new ways of storytelling. They are fully aware that the VDM is an AI entity that continuously generates story prompts while taking into account players' text inputs.

While AI has made huge advancements in enabling artificial entities to mimic human abilities, inconsistencies in the quality of the generated prompts affect players' feelings of competence and challenge with the game. New story elements that suddenly appear in the generated story prompts without proper introduction and context can disrupt the flow and sensory immersion of the participants and do not give participants enough time to build the current story elements. Lastly, the required computing power to run the GPT2 transformer model and the BERTScore model caused delays in the VDM's response time. These delays can disrupt the players' thoughts and momentum and affect their tension with the events happening in the game, thereby reducing their immersion in the fictional game world.

User Interface. Playing D&D heavily relies on the use of words to describe events taking place in the story world. A minimalistic UI such as the one employed by Fable Reborn should easily support player-to-player and player-to-VDM text-based interaction. While players who are well-versed in the mechanics of D&D found the UI to be aesthetically pleasing, other modes of interaction available in tabletop D&D such as gestures and tone of voice are currently not supported.

6 CONCLUSION

Fable Reborn provides an alternative platform for tabletop D&D players to role-play their characters with an AI-powered virtual DM. Players found VDM to be capable of producing engaging and believable fantasy stories but lacks consistency in the quality of its generated stories. The formulated responses are aligned with the human players' story text inputs while occasionally introducing random and unrelated new elements.

However, participants also expressed their sentiment that Fable Reborn will probably not truly replace a human game master in the tabletop D&D game. The human DM is someone who can communicate back with the players on their intentions on the campaign rather than the virtual DM who just weaves the story

around the current story and the players' text inputs. Human DMs can also steer the players to the campaign's goal. Future work should consider empowering the virtual DM with the same abilities.

REFERENCES

- [1] Aaron Bangor, Philip Kortum, and James Miller. 2009. Determining what individual SUS scores mean: Adding an adjective rating scale. *Journal of Usability Studies* 4, 3 (2009).
- [2] John Brooke. 1995. SUS: A 'Quick and Dirty' Usability Scale. In *Usability Evaluation in Industry, 1st Edition*. CRC Press.
- [3] Paul Cairns, Anna Cox, and A. Imran Nordin. 2014. Immersion in Digital Games: Review of Gaming Experience Research. In *Handbook of Digital Games*. <https://doi.org/10.1002/9781118796443.ch12>
- [4] Gordon Calleja. 2011. *In-Game: From Immersion to Incorporation*. MIT Press. <https://doi.org/10.7551/mitpress/8429.001.0001>
- [5] Amal S. Fadak and Osama Khozium. 2021. Dialogue Generation for Digital Interactive Storytelling. *International Journal of Future Generation Communication and Networking* 13, 4 (2021), 4451–4460.
- [6] Wijnand A. IJsselstein, Yvonne De Kort, and Karolien Poels. 2013. *The Game Experience Questionnaire*. Technische Universiteit Eindhoven.
- [7] Wijnand A. IJsselstein, Yvonne De Kort, Karolien Poels, Audrius Jurgelionis, and Francesco Bellotti. 2007. Characterising and Measuring User Experiences in Digital Games. In *International Conference on Advances in Computer Entertainment Technology (ACE '07)*. ACM.
- [8] Orr Group. 2021. Roll20: The Orr Group Industry Report Q4 2020: 8 Million Users Edition! <https://blog.roll20.net/posts/the-orr-group-industry-report-q4-2020-8-million-users-edition/>
- [9] Mark Riedl and Vadim Bulitko. 2012. Interactive Narrative: An Intelligent Systems Approach. *AI Magazine* 34, 1 (2012). <https://doi.org/10.1609/aimag.v34i1.2449>
- [10] Mark Riedl, B. Li, H. Ai, and A. Ram. 2011. Robust and Authorable Multiplayer Storytelling Experiences. In *Proceedings of the Seventh AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE'11)*. 189–194.
- [11] Jonathan P. Rowe, Scott W. McQuiggan, Jennifer L. Robinson, Derrick R. Marcey, and James C. Lester. 2009. STORY EVAL: An Empirical Evaluation Framework for Narrative Generation. In *Proceedings of the AAAI 2009 Spring Symposium*. 103–110.
- [12] Jeff Sauro. 2011. *A Practical Guide to the System Usability Scale: Background, Benchmarks and Best Practices*. Measuring Usability, LLC, Denver.
- [13] Nathan Shank. 2011. *The Narrative Structures of Dungeons and Dragons*. (2011). https://www.academia.edu/3873121/Masters_Thesis_The_Narrative_Structures_of_Dungeons_and_Dragons
- [14] Stanley Yu Galan, Michael Joshua Ramos, Aakov Dy, Yusin Kim, and Ethel Ong. 2017. Towards a Narrative-Based Game Environment for Simulating Business Decisions. In N. Nunes, I. Oakley and V. Nisi (eds): *Interactive Storytelling (ICIDS 2017)*, Vol. LNCS 10690. Springer International Publishing.
- [15] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. BERTScore: Evaluating Text Generation with BERT. <https://doi.org/10.48550/arXiv.1904.09675>